

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0006420713806700977
rotor delay [s]: 0.145400849371
gimbal delay [s]: 0.037953368708
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 4.851501173069026e-17
common-scale FD null norm: 1.0685767737147155e-11

Dimensionless coordinate

log_mass_scale	-0.187662184897
log_second_moment_mode_xx	0.870339025138
log_second_moment_mode_yy	0.52938102128
log_second_moment_mode_zz	0.237215125147
log_second_moment_mode_xy	-0.0375329126012
log_second_moment_mode_xz	0.695400740999
log_second_moment_mode_yz	-0.0847672281319
dimensionless_cog_displacement_x	-0.173270270882
dimensionless_cog_displacement_y	0.00775171195766
dimensionless_cog_displacement_z	-0.293344685056
log_force_effectiveness_1	-0.22536220077
log_force_effectiveness_2	-0.470638545158
log_force_effectiveness_3	-0.493462162788
log_force_effectiveness_4	-0.259810077953

Physical representative

mass [kg]: 1.9492267162555124
CoG body [m]: [-0.03522069 0.00145459 -0.04669069]
force effectiveness: [0.79822705 0.6246033 0.61050905 0.77119804]
inertia [kg m²]:
[0.1105048421, 0.005282153722, -0.004999928271]
[0.005282153722, 0.1702281403, 0.0005767399509]
[-0.004999928271, 0.0005767399509, 0.2793299347]
principal moments [kg m²]: [0.10989187 0.17069154 0.2794795]

Per-bag raw residual-wrench RMS

failure_2_single: [0.16886726 0.26818765 0.45673595 0.05307211 0.14389321 0.12003766]